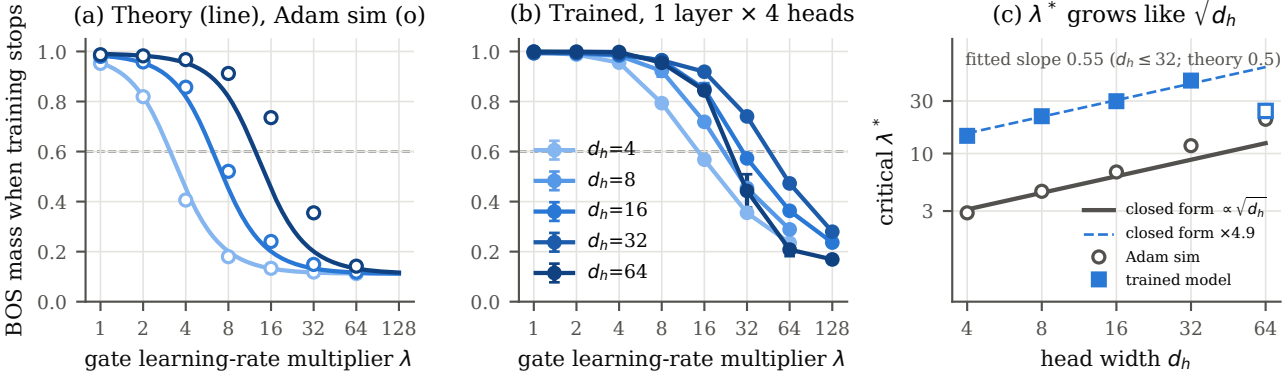
(a) Theory (line), Adam sim (o)
BOS mass when training stops
1.0
0.8
0.6
0.4
0.2
0.0
1
2
4
8
16
32
64
128
gate learning-rate multiplier λ
(b) Trained, 1 layer × 4 heads
d_h=4
d_h=8
d_h=16
d_h=32
d_h=64
gate learning-rate multiplier λ
(c) λ* grows like √d_h
fitted slope 0.55 (d_h ≤ 32; theory 0.5)
critical λ*
30
10
3
closed form ∝ √d_h
closed form ×4.9
Adam sim
trained model
4
8
16
32
64
head width d_h